Aditya Manoj

B.Tech Computer Science & Engineering — AI/ML Systems & Full-Stack Engineer

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Targeted Technical Profile

Systems & Full-Stack Software Engineer (B.Tech CSE) with **2+ years of production experience** engineering **real-time digital audio processing**, **local deep learning inference engines**, **time-series forecasting models**, and **offline-first Progressive Web Applications**. Architect of ALFRED Mark-IX (dual-stage acoustic wake-word neural networks, lock-free ring buffers, and 512-D biometric speaker verification across 125K+ LoC) and commercial systems spanning Python ML pipelines, high-framerate HTML5/WebGL mathematical visualizers, and multi-backend cloud platforms. Proven track record deploying resilient architectures with strict data durability across **Python (NumPy, PyTorch, ONNX)**, **TypeScript/React**, and **Cloud/DB Systems (Firebase, Supabase, IndexedDB, AWS)**.

Technical Competencies & Engineering Stack

- **AI/ML Systems & Signal Processing:** PyTorch (AdamW, LayerNorm, ONNX Dynamic Export), ONNX Runtime, OpenWakeWord, Faster-Whisper (CTranslate2 INT8), CAM++ Biometric Speaker Verification (512-D Embeddings), 16 kHz PCM Windowing, Circular Ring Buffers, Time-Series Forecasting (Walk-Forward Backtesting, Feature Matrix Engineering).
- **Full-Stack & Frontend Engineering:** React 18/19, TypeScript, JavaScript (ES Modules), Vite, HTML5 Canvas API, WebGL/Three.js, PWA (Service Workers, CacheStorage, Web Push), Static Site Generation (SSG), Tailwind CSS, PyQt6 (Tactical HUD).
- **Distributed Storage & Cloud Systems:** Cloud Firestore (NoSQL, Rules Unit Testing), Supabase (PostgreSQL), IndexedDB (Dexie.js), Cloudflare Workers, WebAuthn (Passkeys Biometrics), Multi-Tier Backend Adapters, AWS/GCP Edge Delivery, Git/GitHub Actions CI/CD.

Professional Experience

Lead Systems Engineer – GLIA Event OS *Jun 2025 – Nov 2025*
Rajiv Gandhi Centre for Biotechnology (BRIC-RGCB) – glia-app-a6d84.web.app *Thiruvananthapuram*

- Architected and deployed **GLIA** (glia-app-a6d84.web.app, previously hosted at `ian2025.in`), a mission-critical offline-first Event OS & PWA for the IAN-2025 International Neuroscience Symposium, maintaining **100% operational uptime** across delegates from MIT, Cambridge, and RIKEN.
- Orchestrated event telemetry processing **1.2M+ atomic Firestore operations** and 2,275+ cryptographic QR coupon validations with bi-directional reactive synchronization and zero data drift under **Dr. Jackson James** at BRIC-RGCB.

Product Development Engineer *Jan 2025 – Aug 2025*
Lascade – travelanimator.com *Kochi, Kerala*

- Developed full-stack mobile systems and 3D route animation pipelines for **Travel Animator** (travelanimator.com), serving **1,000,000+ creators and travelers** across iOS and Android.
- Engineered high-performance interactive map plotting (Google Maps API), custom 3D vehicle coordinate rendering, and hardware-accelerated HD/4K video export modules.

AI/ML Engineering Intern *Jan 2024 – Jun 2024*
ICT Academy of Kerala *Thiruvananthapuram*

- Built Python data pre-processing and signal augmentation pipelines using NumPy and OpenCV, boosting model classification accuracy by **10%** across supervised convolutional deep neural networks.

Full-Stack Development Intern *Jul 2024 – Oct 2024*
ICT Academy of Kerala *Thiruvananthapuram*

- Engineered responsive frontend interfaces and RESTful backend APIs, implementing client-side state synchronization, component caching, and relational database schema optimizations.

Education

B.Tech in Computer Science & Engineering *2021 – 2025*
Jyothi Engineering College (APJ Abdul Kalam Technological University) *Thrissur, Kerala*

- **Core Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems (DBMS), Machine Learning, Digital Signal Processing fundamentals, Object-Oriented Systems (Python/C++), Computer Networks.

Flagship Engineering Systems & Production Deployments

ALFRED Mark-IX – Autonomous Multimodal AI Desktop Assistant — https://github.com/AdityaManojA/ALFRED-MK *2024 – Present*

- Directed architecture across a 125,000+ LoC Python systems codebase with 5 contributors (6 merged PRs, avg. 321 LoC/PR) and an automated test suite of 1,705 tests and 4,946 assertions (60.8% line coverage, 100% valid AST syntax).
- Architected continuous 16 kHz mono stream pipeline with **SoundDevice/NumPy** (5.0 s ring buffer, RMS gating), dual-gate ONNX ensemble (512-D CAM++ voice biometrics), and 60 FPS PyQt6 tactical HUD (22 ms warm latency).

The Whisky Business – Commercial Suite & ML Demand Forecaster — https://thewhiskybusiness.in *2026*

- Built operations platform, customer loyalty PWA, and Python **walk-forward rolling backtest forecasting engine** across 30/56-day sliding lookback windows, projecting 14-day demand with dynamic confidence interval bands ($M \ldots Z$ polygon shading).
- Formulated cultural calendar model weighting festival shocks (**+115% Durga Ashtami**, **+92% Mahanavami**, **+50% Thiruvonam**) and 7×12 heatmap (peak 7.0× Sunday surges); captured **#1 and #3 Google rankings** and **Google AI Overviews** (34% CTR).

ThreadPath Collective – Sustainable E-Commerce Storefront — https://threadpathcollective.ca *2026*

- Engineered a full-featured e-commerce storefront for a sustainable Canadian streetwear brand, integrating high-throughput product catalogs, faceted category filtering, and fluid Framer Motion micro-interactions.
- Architected search metadata and semantic markup to secure the **#1 organic Google ranking** and featured **Google AI Overview**; deployed across hybrid AWS/GCP cloud edge networks with sub-second page transitions.

Levern – Gamified Quest Engine & Habit Tracker PWA — https://getlevern.web.app *2026*

- Shipped and maintain a mobile-first productivity PWA serving **55 Daily Active Users (DAU)** with zero framework dependencies in pure **vanilla ES modules** and semantic CSS, delivering 100% offline persistence.
- Developed in-memory natural-language schedule parser (e.g., "water plants every other day at 6pm"), seasonal leaderboard rankings, interactive companion sprite physics, and automated 5-minute Web Push alerts via GitHub Actions.

TerraCove – Multi-Backend Freelance Workspace (MVP) — https://terracove.web.app *2026*

- Architected a freelance operations MVP managing clients, deliverables, invoicing, and approvals with pre-launch sandboxed workspace initialization.
- Engineered an abstract, switchable multi-backend data adapter supporting runtime routing between **Cloud Firestore (NoSQL)**, **Supabase (PostgreSQL)**, and local offline state; built gesture scheduling and client approval portals.

IndZita Biotech – Non-Invasive Diagnostics & Bio-Automation Platform — https://indzitabio.web.app *2026*

- Architected official web platform and clinical showcase for IndZita Biotech, backed by an **INR 7 Crore (approx. $840,000 USD) MAHA MedTech grant** from **ANRF, ICMR, and Gates Foundation** with **BRIC-RGCB & Karkinos Healthcare**.
- Showcases non-invasive cervical cancer diagnostics using blood-based **piRNA biomarkers** and **nanopore single-molecule sensing**, alongside plate-based **organoid bioreactors** (PTFE alloy, 720+ run hours) and React 18 / Vite generative ASCII genomic canvas.

Endogenous App – Government Rural & Heritage Tourism Platform — *Public Sector Initiative* *2025*

- Architected specialized mobile and web platform for the government-backed Endogenous Tourism Project, delivering interactive geospatial mapping, cultural heritage trails, and local artisan directories.
- Engineered low-bandwidth offline caching protocols via Service Workers, multi-lingual localization, and secure digital onboarding pipelines to empower rural community tourism enterprises.

Pixel Spaces – 3D Visualisation Studio Platform & SSG Engine — https://pixelspaces.web.app *2026*

- Built architectural studio web platform featuring a bespoke **Static Site Generation (SSG)** pipeline pre-rendering static HTML for all routes with client-side hydration, dynamic Open Graph tags, and Schema.org JSON-LD.
- Engineered automated multi-resolution asset pipeline with **Node.js and Sharp** (WebP tiers 960w/1920w), 3D model processing pipeline (`obj-to-glb`), and interactive HTML5 Canvas "Pixel-Resolve" hero stage.

Institutional Alignment & Certifications

- **BRIC-RGCB Track Record:** Direct on-site experience at BRIC-RGCB Thiruvananthapuram; institutional clearance and high-security deployment familiarity.
- **Meta Full Stack Developer Specialization:** Meta/Coursera (Certified in modern frontend/backend patterns, CI/CD, and production deployment).
- **Academic & Technical Commendations:** Awarded technical leadership recognition supporting biomedical symposium computing infrastructure.